Test Series Code

14

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 19/04/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

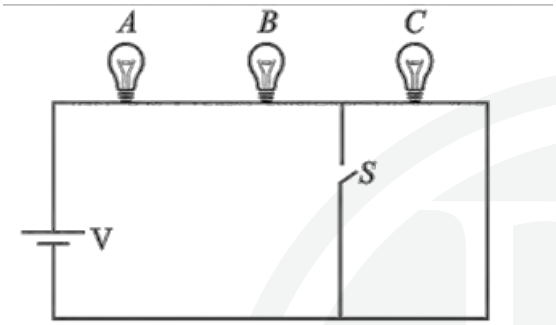
Student's Signature: _____ **Invigilator's Signature:** _____

1. A particle of mass m starts from rest and moves in a straight line such that its velocity varies with displacement x as $v = \alpha\sqrt{x}$. The work done by all forces acting on the particle in the first t seconds is:

(1) $\frac{m\alpha^4 t^2}{4}$ (2) $\frac{m\alpha^4 t^2}{8}$

(3) $\frac{m\alpha^2 t^2}{2}$ (4) $\frac{m\alpha^2 t^4}{8}$

2. Three identical bulbs A, B and C are connected in a circuit as shown. What will happen to the brightness of bulbs, if the switch S is closed?



(1) Brightness of bulbs A and B will decrease.
(2) Brightness of bulbs A and B will increase.
(3) Brightness of bulb C will increase.
(4) Brightness of bulb A will decrease and B will increase.

3. Consider a vertical spring with a spring constant k hanging from a fixed support. A block of mass m is attached to its lower end and allowed to come to rest at its equilibrium position. While the block is at rest, an external constant downward force of magnitude $F = mg$ is suddenly applied to it. After the force is applied, what will be the maximum total extension of the spring, measured from its original unstretched length? (g is acceleration due to gravity)

(1) $\frac{mg}{k}$ (2) $\frac{2mg}{k}$

(3) $\frac{3mg}{k}$ (4) $\frac{4mg}{k}$

4. A satellite revolves in a circular orbit at a height $h = R$ above the surface of the Earth (where R is Earth's radius). If it is shifted to a new circular orbit at a height $h = 3R$, its new orbital speed becomes:

(1) Half of its original speed
(2) $\frac{1}{\sqrt{2}}$ times its original speed
(3) $\frac{1}{4}$ of its original speed
(4) $\frac{1}{\sqrt{3}}$ times its original speed

5. AC voltage of $V = 220\sqrt{2}\sin\left(100\pi t + \frac{\pi}{2}\right)$ volt is applied across a DC voltmeter; its reading will be:

(1) $220\sqrt{2}$ V (2) $\sqrt{2}$ V
(3) 220 V (4) zero

6. A particle of mass m is projected from the ground with an initial speed v_0 at an angle of 60° to the horizontal. What is the magnitude of its angular momentum about its highest point at the instant it lands on the ground? (g is acceleration due to gravity)

(1) $\frac{\sqrt{3}mv_0^3}{16g}$ (2) $\frac{3mv_0^3}{16g}$

(3) $\frac{mv_0^3}{8g}$ (4) $\frac{3mv_0^3}{8g}$

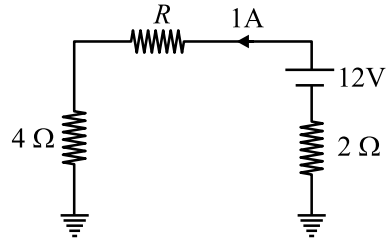
7. Water (density $\rho = 1000\ \text{kg/m}^3$) flows steadily through a horizontal pipe of varying cross-section. The cross-sectional area at the wider section Point (A) is $40\ \text{cm}^2$ and at the narrower section Point (B) is $10\ \text{cm}^2$. If the pressure difference between Point A and Point B is 7500 Pa, what is the speed of the water at Point A?

(1) 0.5 m/s
(2) 1.0 m/s
(3) 2.0 m/s
(4) 4.0 m/s

8. A Carnot engine operates between temperatures T_1 (source) and T_2 (sink). If the source temperature is increased by 20% and the sink temperature is decreased by 20%, the new efficiency becomes $\frac{5}{3}$ times the original efficiency. The original efficiency of the engine was:

(1) 25% (2) 33.3%
(3) 40% (4) 50%

9. Value of the resistance R in the figure is:

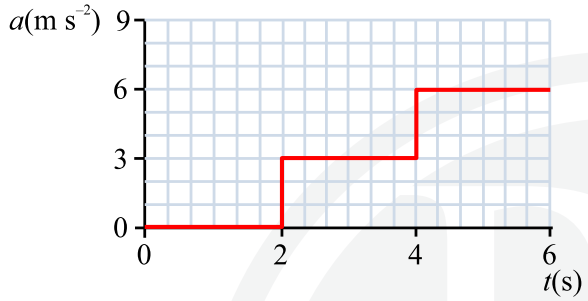


(1) 6 Ω (2) 8 Ω
(3) 10 Ω (4) 12 Ω

10. The maximum kinetic energy of a photoelectron emitted from a metal surface with a work function f, due to incident light of frequency n, is E. If the frequency of the incident light is doubled, what will be the new maximum kinetic energy of the emitted photoelectrons?

(1) $2E$ (2) $2E - f$
(3) $2E + f$ (4) $2E + 2f$

11. A particle moves in a straight line with an initial velocity of 2.00 m/s. From the acceleration-time graph provided, find the velocity of the particle at $t = 6$ s.



(1) 18.0 m/s (2) 20.0 m/s
(3) 9.0 m/s (4) 11.0 m/s

12. Given below are two statements:

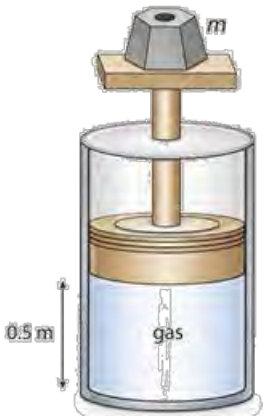
Statement I: Rutherford's alpha-particle scattering experiment led to the conclusion that most of the mass and all of the positive charge of an atom are concentrated in a tiny central core called the nucleus.

Statement II: Rutherford's atomic model successfully explained the electrical stability of the atom and the discrete line spectrum of hydrogen.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

13. A cylinder is placed in a vacuum and has a movable, frictionless piston at the top. An ideal gas is kept inside the cylinder. The volume of the gas is 0.050 m^3. A weight on top of the piston has a mass of 10.0 kg. Assuming the mass of the piston itself is negligible, what is the pressure of the gas at equilibrium? (Take $g = 9.8$ m/s^2)



(1) 98 Pa (2) 980 Pa
(3) 101,300 Pa (4) 102,280 Pa

14. A uniform conducting rod CD of length $3L$ is placed in a uniform magnetic field B pointing into the plane. The rod is pivoted at a point P located at a distance L from one end (end C) and rotated with an angular velocity ω in a plane perpendicular to field B. The induced potential difference between its ends C and D is:

(1) $1.5B\omega L^2$ (2) $2B\omega L^2$
(3) $2.5B\omega L^2$ (4) $4B\omega L^2$

15. The electric field component of a plane electromagnetic wave is given by $E_y = 50\sin(10^8 t - 0.5x)$ V/m, where x is in meters and t is in seconds. The refractive index of the medium through which this wave is traveling is:

(1) 1.0 (2) 1.5
(3) 2.0 (4) 2.5

16. A symmetric biconvex lens made of glass ($n = 1.5$) has a focal length of 20 cm. The lens is cut into two identical halves along a plane parallel to its principal axis. If one of these halves is used to form an image of an object placed at a distance of 30 cm from the lens, the image will be formed at:

(1) 30 cm from the lens, virtual
(2) 60 cm from the lens, real
(3) 30 cm from the lens, real
(4) 60 cm from the lens, virtual

17. The de Broglie wavelength of an electron accelerated from rest through a potential difference V is λ. If the accelerating potential difference is increased to 4V, its new de Broglie wavelength will be:

(1) 2λ (2) λ
(3) $\frac{\lambda}{2}$ (4) $\frac{\lambda}{4}$

18. An electron (charge e) revolves in a uniform circular orbit of radius r with a constant speed v. The magnitude of the magnetic field produced at the center of the orbit is: (μ_0 is permeability of free space)

(1) $\frac{\mu_0 ev}{2\pi r^2}$ (2) $\frac{\mu_0 ev}{4\pi r^2}$

(3) $\frac{\mu_0 ev}{2r}$ (4) $\frac{\mu_0 ev}{4r^2}$

19. A point charge $+q$ is placed at a distance d from the center of a small cube of side length a (where $d > 2a$). This small cube is located entirely inside a large hollow cubical box of side length L (where $L >> a$). What is the total electric flux passing through the surface of the small cube?

(1) $\frac{q}{\epsilon_0}$

(2) Zero

(3) $\frac{q}{6\epsilon_0}$

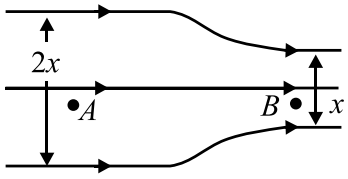
(4) Cannot be determined without the exact position of the charge relative to the small cube.

20. The dimensional formula of the product $F\epsilon_0$, where F is the force and ϵ_0 is the permittivity of free space, is:

(1) $[M^0L^{-2}T^2A^2]$ (2) $[M^2L^4T^{-6}A^{-2}]$

(3) $[M^0L^2T^{-2}A^{-2}]$ (4) $[M^0L^{-2}TA]$

21. The figure shows the pattern of electric field lines in a certain region of space. Let E_A and E_B be the magnitudes of the electric field at points A and B respectively. Based on the diagram, we conclude:



(1) $E_A < E_B$ (2) $E_A > E_B$

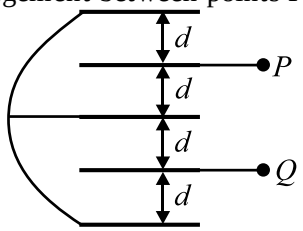
(3) $E_A = E_B$ (4) data insufficient

22. A swimmer whose speed in still water is $2v$ wants to cross a river of width W flowing with speed v. The ratio of the time taken to cross the river along the shortest possible path to the time taken to cross it in the shortest possible time is:

(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$

(3) 2 (4) $\frac{1}{2}$

23. Five identical metallic plates are arranged as shown in the figure. Each plate has a surface area 'A' on one side, and the distance between any two adjacent plates is 'd'. What is the equivalent capacitance of this arrangement between points P and Q?



(1) $\frac{2}{3}\frac{\epsilon_0 A}{d}$ (2) $\frac{3}{2}\frac{\epsilon_0 A}{d}$

(3) $\frac{\epsilon_0 A}{d}$ (4) $3\frac{\epsilon_0 A}{d}$

24. An ideal diatomic gas undergoes a thermodynamic process defined by $P \propto V^2$. The molar heat capacity for this process is:

(1) $\frac{17R}{6}$ (2) $\frac{5R}{2}$

(3) $\frac{13R}{6}$ (4) $\frac{19R}{6}$

25. The magnetic flux through a coil of resistance $6\,\Omega$ varies with time as $\phi(t) = \frac{t^3}{3} - \frac{6}{2}t^2 + 9t + 2$ (in weber). At what time (in sec) is the current in the circuit zero?

(1) $t = 4$ (2) $t = 2$

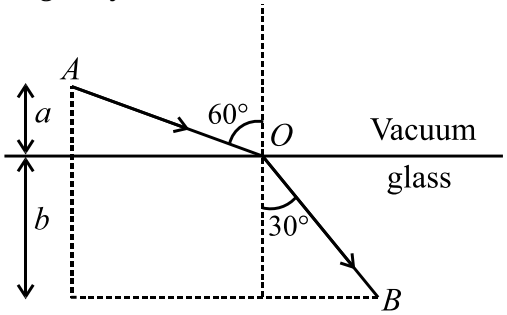
(3) $t = 3$ (4) $t = 6$

26. On the axis of a short magnetic dipole, the magnetic field at a point r = 10 cm from its centre is $B = 2.0 \times 10^{-4}$ T . The magnetic moment of the dipole is (use $\frac{\mu_0}{4\pi} = 10^{-7}$ Tm A^{-1}):

(1) 0.5 Am^2 (2) 1.0 Am^2

(3) 2.0 Am^2 (4) 4.0 Am^2

27. A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is:

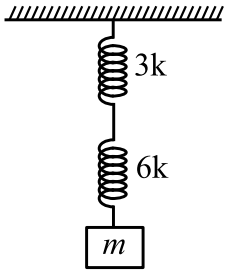


(1) $2a + \frac{2b}{\sqrt{3}}$ (2) $2a + \frac{2b}{3}$

(3) $\frac{2\sqrt{3}}{a} + 2b$ (4) $2a + 2b$

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28. A block of mass m is suspended using two massless springs of spring constant 3k and 6k connected in series. If the block is displaced vertically and released, it executes SHM with a time period of:



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $2\pi\sqrt{\frac{m}{2k}}$

(3) $2\pi\sqrt{\frac{2m}{k}}$ (4) $\pi\sqrt{\frac{m}{k}}$

29. Match the physical system described in **List-I** with its corresponding wave frequency formula in **List-II**. (Here: L = length, v = speed of sound, T = tension, μ = linear mass density)

List-I (Physical System)		**List-II (Frequency Formula)**	
(A)	The fundamental frequency of a stretched string fixed at both ends.	(I)	$\frac{3v}{4L}$
(B)	The fundamental frequency of an air column in a pipe closed at one end.	(II)	$\frac{v}{4L}$
(C)	The first overtone of an air column in a pipe open at both ends.	(III)	$\frac{1}{2L}\sqrt{\frac{T}{\mu}}$
(D)	The first overtone of an air column in a pipe closed at one end.	(IV)	$\frac{v}{L}$

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-IV, C-III, D-II
(4) A-IV, B-I, C-III, D-II

30. When a pure (intrinsic) silicon crystal is doped with a small amount of a trivalent impurity like aluminium, which of the following accurately describes the resulting material?

(1) Its overall electrical resistance decreases compared to the pure crystal.
(2) It becomes a p-type extrinsic semiconductor.
(3) Holes become the majority charge carriers, outnumbering free electrons.
(4) All of these are correct.

31. A potential barrier of 0.50 V exists across a P-N junction. If the depletion region is 5.0×10^{-7} m wide, the intensity of the electric field in this region is:

(1) 1.0×10^{6} V/m
(2) 1.0×10^{5} V/m
(3) 2.0×10^{5} V/m
(4) 2.0×10^{6} V/m

32. A copper calorimeter has a water equivalent of 9 g and contains 21 g of ice at 0°C. Steam at 100°C is passed into the system until the final steady temperature of the mixture is 10°C. The mass (m) of the steam added is:
(Given: $L_f = 80$ cal/g, $L_v = 540$ cal/g, $c_w = 1$ cal/g°C)

(1) 2.1 g (2) 3.14 g
(3) 4.2 g (4) 5.5 g

33. In an interference experiment, two coherent light sources have an intensity ratio of 9:1. What will be the value of the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ in the resulting interference pattern?

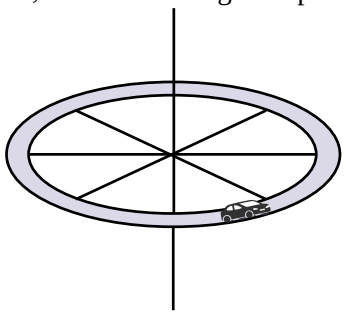
(1) $\frac{3}{10}$
(2) $\frac{3}{5}$
(3) $\frac{3}{100}$
(4) $\frac{3}{50}$

34. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): The density of all atomic nuclei is approximately the same, regardless of their mass number.
Reason (R): The radius (R) of a nucleus depends on its mass number (A) according to the relation $R = R_0 A^{1/3}$, where R_0 is a constant.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

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35. A battery-driven toy car of mass 0.18 kg is placed on a circular track that is part of a horizontal ring. The ring has a moment of inertia of 0.20 kg m^2 and a radius of 0.50 m, and it can rotate about a central vertical axis without friction. If the car is started and its speed relative to the ground is measured to be 0.80 m s^{-1}, what is the angular speed of the ring?



(1) 0.18 rad s^{-1} (2) 0.36 rad s^{-1}
(3) 0.72 rad s^{-1} (4) 1.60 rad s^{-1}

36. A block is held at rest on a rough inclined plane with an angle of inclination of 60°. If the normal force exerted by the plane on the block is 15 N, what is the mass of the block? (Take g = 10 m/s^2)
(1) 1.5 kg (2) 3 kg
(3) $1.5\sqrt{3}$ kg (4) 6 kg

37. Stress - strain curves for two different materials, *X* and *Y* are drawn.
For material *X*, the yield Strength point and the Fracture point are far apart.
For material *Y*, the yield Strength point and the Fracture point are very close to each other.
Which of the following is **correct** regarding their nature?
(1) *X* is Brittle; *Y* is Ductile
(2) *X* is Ductile; *Y* is Brittle
(3) Both are Ductile
(4) Both are Brittle

38. A solid cylinder of mass 2 kg and radius 0.5 m is free to rotate about its fixed geometric axis. A tangential force of 12 N is applied for 4 seconds. The number of revolutions made by the cylinder starting from rest, in this time interval is (Take π = 3): (Assume cylinder is initially at rest)
(1) 8 (2) 16
(3) 32 (4) 64

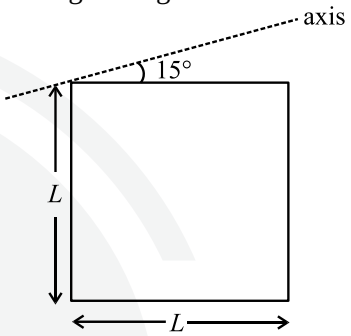
39. The energy of a hydrogen like atom in its ground state is –122.4 eV. It is:
(1) He^+ (2) Li^{2+}
(3) Be^{3+} (4) None of these

40. When 10V, DC is applied across a coil, current through it is 2.5 A. If 10V, 50 Hz A.C. is applied current reduces to 2 A. Calculate reactance of the coil.
(1) 1 Ω (2) 5 Ω
(3) 3 Ω (4) 4 Ω

41. A hypothetical planet has a mass 8 times the mass of Earth, and its uniform density is also 8 times the density of Earth. The ratio of the escape velocity on this planet to the escape velocity on Earth is:
(1) 2 (2) $2\sqrt{2}$
(3) 4 (4) 8

42. Two vibrating tuning forks produce waves given by $y_1 = (2\sin 500\pi t)\,\text{m}, y_2 = (1\sin 506\pi t)\,\text{m}$. Number of beats produced per minute is:
(1) 3 (2) 180
(3) 120 (4) 67

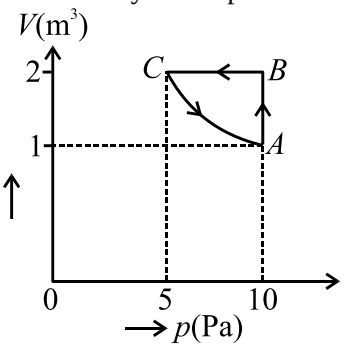
43. A square plate of mass *M* and edge *L* is shown in figure. The moment of inertia of the plate about the axis in the plane of plate passing through one of its vertex making an angle 15° from horizontal is:



(1) $\frac{ML^2}{12}$ (2) $\frac{11ML^2}{24}$
(3) $\frac{7ML^2}{12}$ (4) $\frac{ML^2}{6}$

44. Which of the following statements is **wrong** in case of Maxwell's law of distribution of speeds?
(1) Most probable speed increases with increase in temperature of gas.
(2) Most probable speed is less than mean speed of gas.
(3) Speed of molecules of gas is always less than root mean square speed.
(4) The speed of molecules of a gas may be any value between zero and infinity.

45. An ideal gas is taken through the cycle $A \to B \to C \to A$, as shown in figure. If the net heat supplied to the gas in cycle is 5J, work done by the gas in the unknown thermodynamic process $C \to A$.



(1) –5 J (2) –10 J
(3) 15 J (4) –20 J

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46. 1 mole of an ideal gas A ($C_{V,m} = 3R$) and 2 mole of an ideal gas B are $\left(C_{V,m} = \frac{3}{2}R\right)$ taken in a container and expanded reversible and adiabatically from 1 litre to 4 litre starting from initial temperature of 320 K. ΔE or ΔU for the process is:
(1) –240 R (2) 240 R
(3) 480 R (4) –960 R

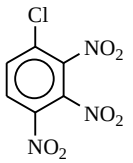
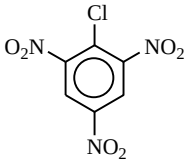
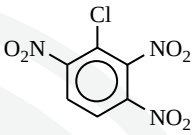
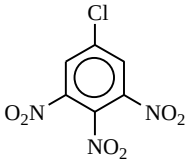
47. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;
Assertion A: Diamond is covalent, yet it has high melting point.
Reason R: Diamond has a three-dimensional network involving strong C–C bonds, which are very difficult to break.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

48. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;
Assertion A: For strong electrolytes, conductivity decreases with decrease in concentration.
Reason R: On dilution total number of ions per unit volume of solution decreases.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

49. If radiation corresponding to second line of "Balmer series" of Li^{2+} ion, knocked out electron from first excited state of H-atom, then kinetic energy of ejected electron would be:
(1) 2.55 eV (2) 4.25 eV
(3) 11.25 eV (4) 19.55 eV

50. 0.5 mole each of glucose and fructose are dissolved in 500 g of water. The freezing point of the resulting solution will be:
(Given: K_f for water = 1.86 K kg mol^{-1})
(1) –3.72°C (2) –1.86°C
(3) –2.48°C (4) –0.82°C

51. Which of the following coordination compounds is used as catalyst for hydrogenation of alkenes?
(1) trans-$[Pt(NH_3)_2Cl_2]$
(2) $[(Ph_3P)_3RhCl]$
(3) $K_2[PdCl_4]$
(4) cis-$[Pt(NH_3)_2Cl_2]$

52. Which of the following compound undergo ArS_N reaction most easily?
(1) 
(2) 
(3) 
(4) 

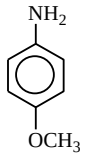
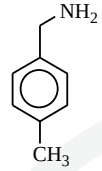
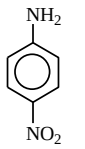
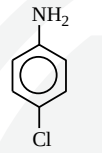
53. Which electronic level would allow the hydrogen atom to absorb a photon but **not** to emit a photon?
(1) 1s
(2) $4p_y$
(3) 2s
(4) $3d_{yz}$

54. The major product of the following reaction is:
$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2OH \xrightarrow{HBr}$
(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CH_3$
(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-CH_2Br$
(3) $CH_3-\underset{\underset{CH_2Br}{|}}{CH}-CH_2-CH_3$
(4) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{Br}{|}}{C}}-CH_2-CH_3$

55. Propanone on treatment with $Ba(OH)_2$ followed by heating give:
(1) 4-Methylpent-4-en-2-one
(2) Hex-3-en-2-one
(3) Hex-4-en-3-one
(4) 4-Methylpent-3-en-2-one

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56. Given below are two statements:
Statement I: A system at higher temperature has more randomness in it than one at lower temperature.
Statement II: When a system is in equilibrium, the entropy is maximum.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

57. Most basic amine among the following is:
(1) 
(2) 
(3) 
(4) 

58. If the solubility of $Zn(OH)_2$ in water is S mol L^{-1} then the solubility of $Zn(OH)_2$ in a buffer solution of pH = 12 will be:
(1) $4 \times 10^4 S^3$ M (2) $2 \times 10^3 S^3$ M
(3) $4 \times 10^2 S^3$ M (4) $2 \times 10^4 S^4$ M

59. The **correct** statement among the following is:
(1) Gluconic acid is a partial oxidation product of glucose
(2) Gluconic acid can form acetal structure
(3) Gluconic acid is a dicarboxylic acid
(4) Gluconic acid is obtained by oxidation of glucose with HNO_3

60. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Molecule obeying octet rule	(I)	BF_3
(B)	Molecule with expanded octet with no unpaired e^-	(II)	NO
(C)	Molecule with incomplete octet and odd electron	(III)	SCl_2
(D)	Molecule with trigonal planar shape	(IV)	PCl_5

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-III, B-IV, C-II, D-I
(3) A-I, B-III, C-IV, D-II
(4) A-II, B-IV, C-III, D-I

61. Among 3d series elements, enthalpy of atomisation is lowest for:
(1) Mn (2) V
(3) Sc (4) Zn

62. For the elementary reaction $2X + Y \rightarrow X_2Y$, having $k = 1 \times 10^{-4}$ mol^{-2} L^2 s^{-1}, $[X_0] = 0.1$ M and $[Y_0] = 0.2$ M, the rate of reaction after [X] is reduced to 0.08 M, is:
(1) 1.6×10^{-7} Ms^{-1} (2) 5.1×10^{-9} Ms^{-1}
(3) 1.216×10^{-7} Ms^{-1} (4) 1.2×10^{-6} Ms^{-1}

63. Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?
(1) 30 mL HCl and 30 mL NaOH
(2) 30 mL CH_3COOH 30 mL NaOH
(3) 30 mL CH_3COOH 30 mL NH_4OH
(4) 30 mL HCl and 30 mL NH_4OH

64. The ratio of sp^2 hybridised and sp^3 hybridised carbon atom in σ-complex of benzene is:
(1) 6 (2) 5
(3) 4 (4) 3

65. The concentration of an oxalic acid solution is 'x' mol $litre^{-1}$. 40 mL of this solution reacts with 16 mL of 0.05 M acidified $KMnO_4$. What is the pH of 'x' M oxalic acid solution? (Assume that oxalic acid dissociates completely.)
(1) 1.3 (2) 1.699
(3) 1 (4) 2

66. Which of the following will have a meso isomer?
(1) 2,2-dichlorobutane
(2) 1,3-dichloro butene
(3) 1,4-dichlorobutane
(4) 2,3-dichlorobutane

67. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;
Assertion A: Kjeldahl method is applicable to estimate nitrogen in diazo benzene.
Reason R: In Kjeldahl method, nitrogen present in organic compound on treating with concentrated H_2SO_4 gets converted to ammonium sulphate.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

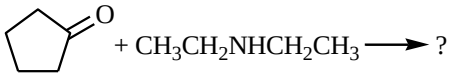
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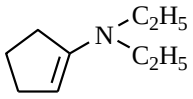
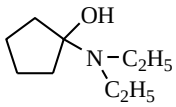
68. The products obtained at cathode and anode during the electrolysis of dilute H_2SO_4 respectively are:

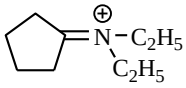
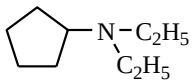
(1) H_2 and $S_2O_8^{2-}$ (2) H_2 and SO_2

(3) O_2 and SO_2 (4) H_2 and O_2

69. What is the major organic product of the following reaction?



(1)  (2) 

(3)  (4) 

70. 5 mole of an ideal gas at temp. T are compressed isothermally from 12 atm. to 24 atm. Calculate the value of 10r. Where,

$$r = \frac{\text{Work done along reversible process}}{\text{Work done along single step irreversible process}}$$

(Given: ln 2 = 0.7)

(1) 7 (2) 10

(3) 5 (4) 6

71. Most stable compound among the following is:

(1) $XY(\Delta_f H^0 = -74\ \text{kJ mol}^{-1})$

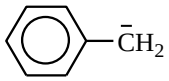
(2) $X_2Y(\Delta_f H^0 = +96\ \text{kJ mol}^{-1})$

(3) $XY_2(\Delta_f H^0 = -180\ \text{kJ mol}^{-1})$

(4) $XY_3(\Delta_f H^0 = +121\ \text{kJ mol}^{-1})$

72. Stability order of given species is:

A. $CH_2=CH-\bar{C}H_2$

B. 

C. $CH_3-CH_2^-$

(1) A > B > C (2) A > C > B

(3) C > A > B (4) B > A > C

73. The major product (A) in the following reaction sequence is:

$$CH_3MgCl \xrightarrow{CO_2} \xrightarrow{H_3O^+} \xrightarrow{LiAlH_4} A$$

(1) CH_3CH_2OH (2) CH_3CH_3

(3) CH_3CHO (4) CH_3OCH_3

74. A 2% solution of substance is isotonic with 3% solution of urea. The molar mass of the substance will be:

(1) 32 g mol^{-1} (2) 54 g mol^{-1}

(3) 40 g mol^{-1} (4) 48 g mol^{-1}

75. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;

Assertion A: Boiling point of n-pentane is more than that of isobutane.

Reason R: Isobutane is more spherical than n-pentane.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

76. The most reactive alkene towards acid catalysed hydration among the following is:

(1) $CH_2=CH_2$ (2) $(CH_3)_2C=CH_2$

(3) $CH_3-CH=CH_2$ (4) $CH_3CH_2CH=CH_2$

77. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;

Assertion A: B_2 is paramagnetic in nature.

Reason R: B_2 contains two unpaired electron in its bonding molecular orbitals.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

78. Given below are two statements:

Statement I: SI unit of density is Kg cm^{-3}.

Statement II: The prefix used for the multiple 10^{-2} is centi.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

79. All the given hydrocarbons give only one monochloro derivative **except**:

(1) Neopentane (2) Ethane

(3) Propane (4) Cyclohexane

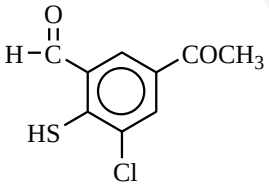
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80. Number of isomeric amines having molecular formula C_3H_9N is:
(1) 2 (2) 3
(3) 4 (4) 5

81. Given below are two statements:
Statement I: All Cu^{2+} halides are known except the iodide.
Statement II: All the f-block elements are radioactive.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

82. For the first order reaction, the slope of graph between $\log\frac{[R]}{[R_0]}$ vs time is:
(1) $-\frac{1}{k}$ (2) $-\frac{2.303}{k}$
(3) $-\frac{k}{2.303}$ (4) $-k$

83. How many moles of Grignard reacts with following compound?

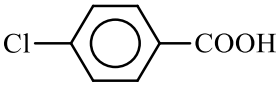


(1) 2 (2) 3
(3) 4 (4) 5

84. Formalin is aq. solution of HCHO
(1) 20% (2) 30%
(3) 40% (4) 10%

85. A vessel at T K contains CO_2 with pressure of 0.4 atm. Some of CO_2 is converted into CO on addition of graphite. The K_p of reaction $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$, will be:
(Given: total pressure at equilibrium is 0.5 atm)
(1) 0.13 atm
(2) 0.28 atm
(3) 0.36 atm
(4) 0.47 atm

86. Number of chiral carbon in β-D-(–)-fructofurannose is:
(1) 6 (2) 5
(3) 4 (4) 3

87. Hell-Volhard-Zelinsky reaction is used for the preparation of:
(1) $CH_3\underset{\underset{Br}{|}}{CH}-COOH$
(2) 
(3) $CH_3-\underset{\underset{I}{|}}{CH}-CH_2COOH$
(4) $CH_3CH_2-\underset{\underset{F}{|}}{CH}-COOH$

88. Match **List-I** with **List-II**.

List-I (Molecule)		List-II (Oxidation state of central atom)	
(A)	S_2F_2	(I)	–2
(B)	N_2H_4	(II)	+5
(C)	H_3PO_3	(III)	+1
(D)	IF_5	(IV)	+3

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-III, C-IV, D-II
(4) A-III, B-I, C-IV, D-II

89. Major product of the following reaction is:
$(CH_3)_3C-Cl + C_2H_5ONa \xrightarrow{\Delta}$
(1) $\begin{matrix}CH_3\\CH_3\end{matrix}\!\!>\!\!=CH_2$
(2) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-O-C_2H_5$
(3) $CH_2=CH_2$
(4) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-C_2H_5$

90. A mixture of NH_4NO_3 and $(NH_4)_2HPO_4$ contain 30.40% mass per cent of nitrogen. What is the mass ratio of the two components in the mixture?
(1) 2 : 1 (2) 1 : 2
(3) 3 : 4 (4) 4 : 1

91. The physico-chemical approach to study and understand living organisms is called;
(1) Chemotaxonomy.
(2) Molecular biology.
(3) Reductionist biology.
(4) Bioinformatics.

92. How many of the following statements regarding prokaryotic cells are correct?
A. The genetic material is not enveloped by a nuclear membrane.
B. Many bacteria have small linear DNA outside the genomic DNA called plasmids.
C. The cell envelope consists of a loosely bound three-layered structure.
D. Ribosomes are associated with the plasma membrane and are of 70S type.
E. Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria.
Choose the **correct** option.
(1) Two only (2) Three only
(3) Four only (4) Five

93. Given below are two statements:
Statement I: DNA is the genetic material in QB bacteriophage.
Statement II: RNA is the genetic material in Tobacco Mosaic Virus.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

94. Match **List-I** with **List-II**.

List-I (Structure)		**List-II (Associated organism)**	
(A)	Pyrenoids	(I)	*Polytrichum*
(B)	Prothallus	(II)	*Marchantia*
(C)	Protonema	(III)	*Volvox*
(D)	Gemmae	(IV)	*Equisetum*

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-IV, D-III

95. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The pyramid of energy is always upright.
Reason (R): When energy flows from a particular trophic level to the next trophic level, some energy is always lost as light at each step.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

96. In a monohybrid cross between a tall pea plant (TT) and a dwarf pea plant (tt), the F_1 progeny is self-pollinated. If a random sample of 1200 plants is obtained in the F_2 generation, what is the expected number of phenotypically tall plants that are also heterozygous?
(1) 300 (2) 600
(3) 900 (4) 1200

97. Which of the following is not a feature of the endodermis in a dicot root?
(1) It comprises a single layer of barrel-shaped cells.
(2) It has intercellular spaces for gaseous exchange.
(3) It has a deposition of waxy material suberin in the form of casparian strips.
(4) It is the innermost layer of the cortex.

98. After double fertilisation in a typical angiosperm, the central cell develops into the _____ and the zygote develops into the _____.
(1) endosperm, embryo
(2) embryo, endosperm
(3) seed, fruit
(4) perisperm, pericarp

99. The 'clot buster' enzyme, Streptokinase that is modified by genetic engineering and used for removing clots from the blood vessels of patients who have undergone myocardial infarction, is produced by:
(1) a fungus. (2) a viroid.
(3) a bacterium. (4) a virus.

100. Arrange the following taxonomic categories related to potato in ascending order of hierarchy.

A. *tuberosum*
B. Plantae
C. Polymoniales
D. Solanaceae
E. *Solanum*

(1) E → D → C → A → B
(2) E → C → D → A → B
(3) B → A → C → D → E
(4) A → E → D → C → B

101. Match **List-I** with **List-II**.

List-I (Population Interaction)		**List-II (Species A; Species B)**	
(A)	Mutualism	(I)	+,0
(B)	Commensalism	(II)	–,0
(C)	Amensalism	(III)	+,+
(D)	Predation	(IV)	+,–

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

102. Identify the **incorrectly** matched pair regarding the steps of glycolysis.

(1)	Glucose → Glucose-6-phosphate	:	ATP utilised
(2)	Fructose-6-phosphate → Fructose 1, 6-bisphosphate	:	H_2O formed
(3)	3-phosphoglyceraldehyde (PGAL) → 1, 3-bisphosphoglycerate (BPGA)	:	NADH + H^+ formed
(4)	1, 3-bisphosphoglycerate (BPGA) → 3-phosphoglyceric acid (PGA)	:	ATP formed

103. What is common between C_3 and C_4 plants?

(1) Primary CO_2 acceptor
(2) Sugar synthesis by Calvin cycle
(3) Presence of Kranz anatomy in leaves
(4) Amount of biomass produced under same conditions

104. In the cell cycle of a drone honey bee, the number of chromosomes at G_1, G_2 and metaphase, respectively, will be;

(1) 32, 32, 32
(2) 16, 32, 16
(3) 16, 16, 16
(4) 32, 16, 32

105. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Diatoms	(I)	Saprophytic protists
(B)	Dinoflagellates	(II)	Some of them cause red tides
(C)	Euglenoids	(III)	'Chief producers' in the oceans
(D)	Slime moulds	(IV)	Have a protein rich layer called pellicle

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-II, C-IV, D-I

106. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Application of ethephon increases yield in cucumbers.

Reason (R): Ethephon promotes female flowers in cucumbers.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

107. Which of the following statements about Mendelian disorders are **correct**?

A. Haemophilia is a sex-linked recessive disease.
B. Sickle-cell anaemia is an autosome-linked recessive trait.
C. Phenylketonuria is the inborn error of metabolism.
D. Colour blindness occurs in about 8 per cent of females and only about 0.4 per cent of males.
E. Thalassemia is a qualitative blood disorder.

Choose the **correct** option.

(1) A, C, E
(2) B, D, E
(3) B, C, D
(4) A, B, C

108. The cells that are present on the chalazal end of a typical embryo sac;
(1) are haploid.
(2) have special thickenings called the filiform apparatus.
(3) constitute the egg apparatus.
(4) degenerate prior to double fertilisation.

109. The first step in decomposition, where detritivores like earthworms break down detritus into smaller particles, is called:
(1) Catabolism.
(2) Leaching.
(3) Humification.
(4) Fragmentation.

110. The Steller's Sea Cow is an example of recent extinction from;
(1) Australia.
(2) Mauritius.
(3) Africa.
(4) Russia.

111. Match **List-I** with **List-II**.

List-I		List-II	
(A)	DNA ligase	(I)	Catalyses transcription
(B)	DNA polymerase	(II)	Joins discontinuously synthesised DNA fragments
(C)	RNA polymerase	(III)	Digests DNA
(D)	DNase	(IV)	Catalyses polymerisation of deoxynucleotides

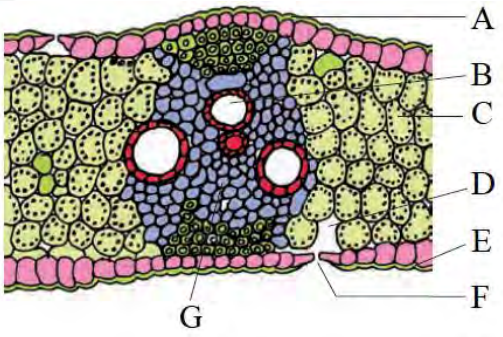
Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-IV, B-II, C-I, D-III

112. How many of the following pairs of plant part and its modification type are **correctly** matched?
(A) Turnip - Storage tap root
(B) Sweet potato - Storage adventitious root
(C) *Rhizophora* - Pneumatophores for respiration
(D) Banyan tree - Stilt roots for support
(E) Maize - Prop roots for support
(1) One only (2) Two only
(3) Three only (4) Four only

113. Given below are two statements:
Statement I: During chemiosmosis in the chloroplast, the primary electron acceptor donates its electrons to the proton carrier.
Statement II: In cyclic photophosphorylation, NADPH + H^+ is not synthesised.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

114. Identify the **incorrect** statement.
(1) Katherine Esau published the "*Anatomy of Seed Plants*".
(2) Carolus Linnaeus published his book titled as "*Systema naturae*".
(3) Panchanan Maheshwari emphasised the need for initiation of work on artificial culture of immature embryos.
(4) Ramdeo Misra was awarded Nobel Prize for establishing the full potential of penicillin as an antibiotic.

115. Shown below is the T.S. of a plant part. Choose the **true** statement about it.



(1) It represents the T.S. of a dicot leaf.
(2) A and E both contain stomata and are covered by cuticle.
(3) B and G constitute the fundamental tissue system.
(4) C, D and F are a part of epidermal tissue system.

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116. Identify the **incorrect** statements about agents of pollination.

A. Majority of plants use abiotic agents for pollination.
B. Pollination by wind is more common amongst abiotic pollinations.
C. Pollination by water is quite rare in flowering plants, limited to about 30 genera, mostly dicotyledons.
D. Flowers pollinated by flies and beetles secrete foul odours to attract these animals.
E. Bees are the dominant biotic pollinating agents.

Choose the **correct** option.

(1) A, C only (2) B, C, D only
(3) A, D, E only (4) C, D only

117. Given below are two statements:

Statement I: In incomplete dominance, the F_1 generation has a phenotype that is in-between the two parents.

Statement II: In the inheritance of flower colour in Snapdragon, the F_2 phenotypic ratio from a cross between red (RR) and white (rr) flowered plants is 1 (Red) : 2 (Pink) : 1 (White).

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

118. A fat molecule breaks down into 8 molecules of acetyl CoA that enters the aerobic respiratory pathway. How many ATP molecules are expected to be synthesised from them via oxidative phosphorylation if they undergo complete breakdown?

(1) 12 (2) 120
(3) 88 (4) 96

119. The fluid mosaic model of the plasma membrane, proposed by Singer and Nicolson, explains that the:

(1) quasi-fluid nature of protein enables the lateral movement of lipids.
(2) quasi-fluid nature of lipid enables the lateral movement of proteins.
(3) membrane is a rigid structure with fixed proteins and lipids.
(4) lipids are arranged in a single layer with proteins embedded in it.

120. The following are all characteristics of Mycoplasma, **except**:

(1) they completely lack a cell wall.
(2) they are the smallest living cells known.
(3) many are pathogenic in animals and plants.
(4) they can survive only in the presence of oxygen.

121. Select the **incorrect** statement regarding bryophytes.

(1) They are also called amphibians of the plant kingdom.
(2) The main plant body of the bryophyte is haploid.
(3) The male sex organ is called antheridium and produces biflagellate antherozoids.
(4) The sporophyte is free-living and independent of the gametophyte.

122. Read the following statements.

A. *Staphylococcus* is used for the production of curd and increases vitamin B_{12} content in it.
B. 'Toddy', a traditional drink, is made by fermenting sap from palms.
C. The large holes in 'Swiss cheese' are due to the production of CO_2 by *Propionibacterium sharmanii*.
D. *Saccharomyces cerevisiae* is used for fermenting malted cereals to produce ethanol.
E. Statins are used as immunosuppressive agents.

The number of **incorrect** statements is;

(1) Five. (2) Four.
(3) Three. (4) Two.

123. In a strain of *E. coli*, the gene '*i*' is mutated such that it cannot produce any repressor protein. If this mutant cell is grown in a medium where lactose is completely absent but has other carbon sources;

(1) the structural genes will not be transcribed because there is no inducer.
(2) the cell will die due to lack of energy.
(3) the genes *z*, *y*, and *a* will be transcribed continuously.
(4) RNA polymerase will bind to the operator.

124. The 'Evil Quartet' is the sobriquet used to describe the four major causes of biodiversity losses. Which of the following is NOT one of them?

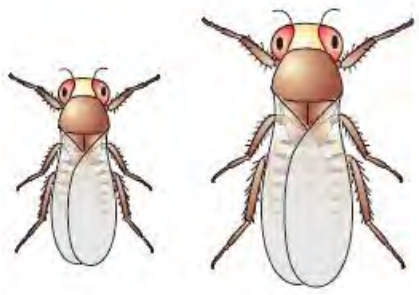
(1) Habitat loss and fragmentation
(2) Over-exploitation
(3) Co-evolution
(4) Alien species invasions

125. Identify the **incorrect** statement about a dicotyledonous stem.
(1) The cortex is differentiated into hypodermis, cortical layers and endodermis.
(2) The pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.
(3) The 'ring' arrangement of vascular bundles is a characteristic of dicot stem.
(4) Pith is small or inconspicuous.

126. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The stage between the two meiotic divisions is generally short-lived.
Reason (R): There is replication of DNA during interkinesis.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

127. Which of the following is not a member of family Poaceae?
(1) Wheat
(2) China rose
(3) Rice
(4) Bamboo

128. If the following pair of fruitfly is crossed, under normal conditions what is the probability of a male progeny being produced with only one sex chromosome?



(1) 0
(2) 0.25
(3) 0.5
(4) 0.75

129. If in an initial population of 375 individuals, 40 died, 56 went to other habitats, 72 were born and 32 came from elsewhere in a period of 1 year; the net change in population;
(1) is 0.
(2) can't be determined.
(3) is increase by 8.
(4) is decrease by 8.

130. In most situations, ___ act(s) as an antagonist to GAs.
(1) Cytokinins
(2) Auxin
(3) Ethylene
(4) Abscisic acid

131. In which type of algae, members reproduce vegetatively by fragmentation, asexually by non-motile spores, and sexually by non-motile gametes?
(1) Green algae
(2) Brown algae
(3) Red algae
(4) Blue-green algae

132. In a grassland ecosystem, the fourth trophic level is occupied by;
(1) goat.
(2) deer.
(3) algae.
(4) man.

133. The most abundant micro-organisms are;
(1) Fungi.
(2) Bacteria.
(3) Protists.
(4) Algae.

134. The process of copying genetic information from one strand of the DNA into RNA is termed as:
(1) Replication.
(2) Translation.
(3) Transcription.
(4) Reverse Transcription.

135. In *Bougainvillea*, the woody, straight, and pointed thorns are modifications of:
(1) Stipules.
(2) Adventitious root.
(3) Axillary buds of stems.
(4) Leaf lamina.

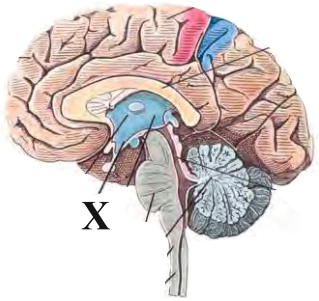
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136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In continuous culture systems, the cells harbouring cloned genes are maintained in their physiologically most active log phase.
Reason: The used medium is drained out from one side while fresh medium is continuously added from the other, thereby preventing the depletion of nutrients and the accumulation of toxic wastes.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Which of the following statements about blood and circulation are **correct**?
A. Rh incompatibility severely affects the first pregnancy.
B. Anti-Rh antibodies after first delivery prevent *erythroblastosis foetalis*.
C. Blood group O lacks anti-A and anti-B antibodies.
D. Thrombin converts fibrinogen into fibrin during clotting.
E. SAN generates 70–75 impulses per minute.
(1) B, D, and E only (2) A, C, and D only
(3) B, C, and E only (4) A, B, D, and E only

138. In a population of 2000 individuals, under Hardy-Weinberg equilibrium, 640 individuals exhibit the heterozygous genotype for a particular trait. Assuming the population is in Hardy-Weinberg equilibrium and the frequency of the dominant allele (p) is less than the frequency of the recessive allele (q), what is the number of individuals expressing the recessive phenotype?
(1) 1280 (2) 640
(3) 1440 (4) 80

139. Consider the sagittal section of the human brain.



What is the function of structure labelled '**X**'?
(1) It integrates information received from the semicircular canals of the ear.
(2) It acts as a bridge interconnecting different regions of the brain.
(3) It contains centres which control body temperature, urge for eating and drinking.
(4) It controls cardiovascular reflexes and gastric secretions.

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	*Ctenoplana*	(I)	Possesses a soft and spongy mantle over a visceral hump.
(B)	*Pinctada*	(II)	Gregarious pest with a chitinous exoskeleton.
(C)	*Taenia*	(III)	Exclusively marine and shows bioluminescence.
(D)	*Locusta*	(IV)	Dorso-ventrally flattened body

Choose the **correct** answer from the options given below:
(1) A-IV, B-II, C-I, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-I, C-III, D-IV

141. Which conversion is initiated by the substance released from activated JG cells?
(1) Conversion of Angiotensin II to Aldosterone
(2) Conversion of Angiotensinogen in blood to Angiotensin I
(3) Conversion of fibrinogen to active fibrin
(4) Conversion of renin to inactive angiotensin

142. How many of the following statements about biomolecules are **correct**?
A. Ribozymes are protein enzymes.
B. All enzymes are damaged above 40°C.
C. Nucleotides have a phosphate group unlike nucleosides.
D. Enzyme–substrate complex formation is essential.
E. Palmitic acid is a 16-carbon saturated fatty acid.
(1) One
(2) Three
(3) Two
(4) Five

143. A 65-year-old man visits a clinic with a significantly weakened cell-mediated immune response. Where is the gland responsible for this, located?
(1) Between the lungs behind the sternum
(2) Attached to the superior pole of each kidney
(3) On the dorsal side of the forebrain
(4) On either side of the trachea

144. Match **List-I** with **List-II** w.r.t respiratory volumes and capacities.

List-I		List-II	
(A)	FRC	(I)	TV + IRV
(B)	IC	(II)	ERV + RV
(C)	EC	(III)	RV + ERV + TV + IRV
(D)	TLC	(IV)	TV + ERV

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

145. Given below are two statements:
Statement I: Placenta provides vascular connection between foetus and uterus.
Statement II: Zygote divides to form blastocyst which is implanted in uterus.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Which of the following contraceptive methods is highly effective, reversible and functions primarily by increasing the phagocytosis of sperms?
(1) Diaphragm
(2) Multiload 375
(3) Saheli
(4) Implants

147. Which of the following infectious agents requires two hosts, human and *Anopheles* mosquito, to complete its life cycle?
(1) *Wuchereria bancrofti*
(2) *Entamoeba histolytica*
(3) *Plasmodium falciparum*
(4) *Salmonella typhi*

148. Identify the **incorrectly** matched pair.
(1) *Bacillus thuringiensis*: Produces inactive protoxins that are solubilised by the alkaline pH of the insect gut.
(2) *cryIAc* and *cryIIAb*: Genes encoding proteins that specifically control the cotton bollworms.
(3) Rosie: The first transgenic cow that produced human protein-enriched milk
(4) RNA interference: A method of cellular defense taking place in all prokaryotes

149. Which of the following event triggers pulling of the actin filaments towards the centre of the 'A' band?
(1) The binding of ATP to the myosin head.
(2) The generation of an action potential in the sarcolemma.
(3) The utilisation of energy from ATP hydrolysis
(4) The return of Calcium ions back to the sarcoplasmic cisternae.

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: If the head of a cockroach is cut off, the insect will still live for as long as one week.
Reason: The head of the cockroach holds only a very small bit of the nervous system, while the majority is situated along the dorsal part of its body.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. In biology**,** describing the exact sequence and position of amino acids in a protein refers to its:
(1) Quaternary structure
(2) Secondary structure
(3) Tertiary structure
(4) Primary structure

152. If the gene controlling Rh factor (D) is autosomal dominant, consider the following:
A. Two Rh-negative parents can have an Rh-positive child.
B. Two Rh-positive parents can have an Rh-negative child.
Choose the **correct** option:
(1) Only A is correct
(2) Only B is correct
(3) Both A and B are correct
(4) Both A and B are incorrect

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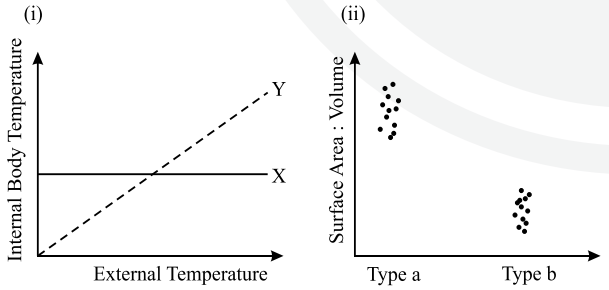
153. Which of the following statements about the evolutionary history of life are **incorrect**?
A. First cellular life originated 2000 mya in water.
B. *Homo erectus* (900 cc brain) likely ate meat.
C. Neanderthals (1400 cc brain) lived in East Africa (100,000–40,000 years ago).
D. *Australopithecines* hunted with stone tools but mainly ate fruits.
E. *Dryopithecus* and *Ramapithecus* (15 mya), with *Ramapithecus* more ape-like.
(1) A, B, and D (2) C and E only
(3) A, C, and E (4) B and D only

154. Which nephron component helps maintain high medullary osmolarity?
(1) PCT (2) Loop of Henle
(3) DCT (4) Bowman's Capsule

155. If an infection causes inflammation in the very thin middle layer of meninges, which layer is primarily affected?
(1) Dura mater (2) Arachnoid
(3) Pia mater (4) Corpus callosum

156. Which part of the female reproductive system serves as the site of fertilization?
(1) Uterus (2) Cervix
(3) Vagina (4) Ampulla of oviduct

157. The figure (i) represents two categories of animals (X and Y) with respect to their response to the external environment. The figure (ii) represents two broad categories of animals (Type a and Type b) with respect to their body surface area to volume ratio.



Which combination most suitably represents the following animals in the order: humming bird; crocodile; frog; polar bear?
(1) Xa; Yb; Ya; Xb
(2) Xb; Ya; Yb; Xa
(3) Ya; Yb; Xa; Xb
(4) Yb; Xb; Xa; Ya

158. Given below are two statements:
Statement I: The DNA fragments separated by agarose gel electrophoresis can be visualized under visible light without any prior staining.
Statement II: If any protein encoding gene is expressed in a heterologous host, it is called a recombinant protein.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. What is the pO_2 in systemic venous blood and systemic arterial blood respectively?
(1) 159 mm Hg and 104 mm Hg
(2) 40 mm Hg and 95 mm Hg
(3) 45 mm Hg and 40 mm Hg
(4) 104 mm Hg and 40 mm Hg

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The hormone glucagon is classified as a hyperglycemic hormone.
Reason: Glucagon acts mainly on hepatocytes to stimulate glycogenolysis and gluconeogenesis.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. Which of the following compounds, typically obtained from a native South American plant have potent stimulating action on CNS producing a sense of euphoria and increased energy.
(1) Heroin (2) Cannabinoids
(3) Cocaine (4) Morphine

162. Given below are two statements:
Statement I: Red muscle fibres are reddish in appearance due to high amount of myoglobin.
Statement II: White muscle fibres depend heavily on aerobic processes for energy.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. Which of the following is **not** typically recognized as an indicator of an improved reproductive health in a society?
(1) Decreased maternal and infant mortality rates
(2) Increased number of medically assisted deliveries
(3) Increased incidences of undetected STIs
(4) Better detection and complete cure of STIs

164. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Gene therapy	(I)	Produces a toxic insecticidal crystalline protein.
(B)	Golden Rice	(II)	Isolation and fusion of naked protoplasts from two different varieties of plants.
(C)	*Bacillus thuringiensis*	(III)	Insertion of genes into an individual's cells and tissues to treat diseases
(D)	Somatic hybridization	(IV)	A genetically modified crop enhanced with Vitamin 'A'.

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-I, B-IV, C-III, D-II

165. Which organ in the alimentary canal of cockroach directly follows the crop and performs the grinding of food?
(1) Mesenteron (2) Gizzard
(3) Hepatic caeca (4) Malpighian tubules

166. Which of the following groups of animals are ammonotelic in nature?
(1) Mammals, terrestrial amphibians, and marine fishes
(2) Reptiles, birds, and land snails
(3) Many bony fishes, aquatic amphibians, and aquatic insects
(4) Cartilaginous fishes, terrestrial insects, and humans

167. When compared to the Earth's crust, which two elements have a relatively higher abundance in any living organism?
(1) Silicon and Sodium
(2) Carbon and Hydrogen
(3) Magnesium and Calcium
(4) Oxygen and Silicon

168. What is the role of the coronary circulation?
(1) Transporting blood from the intestine to the liver before systemic circulation.
(2) Pumping deoxygenated blood from the right ventricle directly to the lungs.
(3) Supplying and collecting blood exclusively to and from the cardiac musculature.
(4) Draining interstitial fluid from tissues back into the major veins.

169. The sudden reversal of polarity at the site of depolarisation on the axon is caused by the:
(1) Active pumping out of 3 Na^+ ions.
(2) Rapid influx of Na^+ ions into the axoplasm.
(3) Rapid efflux of K^+ ions into the extracellular fluid.
(4) Impermeability of the membrane to negatively charged proteins.

170. Which of the following is the **correct** description of interferon?
(1) A synthetic neurotransmitter used to treat neuronal diseases.
(2) A class of proteins secreted by B-lymphocytes that categorically attack and destroy virusinfected cells.
(3) Signalling molecules that interfere with its own synthesis inside a secretory cell.
(4) A class of proteins secreted by virus infected cells and are involved in protecting uninfected cells.

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Homologous organs	(I)	Placental wolf and Tasmanian wolf
(B)	Analogous organs	(II)	Darwin's finches in the Galapagos Islands
(C)	Adaptive radiation	(III)	Sweet potato (root) and potato (stem)
(D)	Convergent evolution	(IV)	Forelimbs of whales, bats, cheetahs, and humans

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-II, D-I

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172. Identify the **correct** True (T) / False (F) combination:

A. Scrotum maintains temperature higher than body temperature for spermatogenesis.

B. Leydig cells lie outside seminiferous tubules and produce androgens.

C. Epididymis continues as vas deferens that loops over the urinary bladder.

(1) A-F, B-F, C-T
(2) A-T, B-T, C-F
(3) A-F, B-T, C-T
(4) A-T, B-F, C-T

173. In order to force bacteria to take up recombinant plasmid DNA during a transformation procedure, the bacterial cells must first be made 'competent'. How is this specific competency chemically achieved?

(1) By incubating the cells with recombinant DNA on ice for 48 hours.
(2) By treating the cells with a specific concentration of a divalent cation, such as calcium.
(3) By bombarding the cells with high-velocity micro-particles of tungsten.
(4) By exposing the bacteria to a continuous bath of the enzyme lysozyme.

174. Which layer of the adrenal cortex is responsible for secreting mineralocorticoids like aldosterone?

(1) Zona reticularis (2) Zona fasciculata
(3) Zona pellucida (4) Zona glomerulosa

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Morphine	(I)	*Erythroxylum coca*
(B)	Cocaine	(II)	*Papaver somniferum*
(C)	Cannabinoids	(III)	Fungi / Microbes
(D)	Penicillin (Antibiotic context)	(IV)	*Cannabis sativa*

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-II, D-I

176. Asthma is primarily characterized by the:

(1) Decrease of the respiratory surface due to alveolar wall damage.
(2) Inflammation of the bronchi and bronchioles.
(3) Proliferation of fibrous tissues
(4) Acute accumulation of pleural fluid

177. Where in the human skeletal system is a classic example of cartilaginous type of joint found?

(1) Between the humerus and the pectoral girdle
(2) Between the flat skull bones forming the cranium
(3) Between the adjacent vertebrae in the vertebral column
(4) Between the carpal and metacarpal of the thumb

178. Which of the following represents the fertile period during a normal 28-day menstrual cycle?

(1) Days 1–5, because menstruation occurs
(2) Days 6–10, due to follicular growth
(3) Days 10–17, due to proximity to ovulation
(4) Days 20–28, due to luteal phase dominance

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The tumor-inducing (Ti) plasmid of *Agrobacterium tumifaciens* has been transformed into a highly effective cloning vector.

Reason: *Agrobacterium tumifaciens* is a pathogen of several dicot plants that naturally delivers 'T-DNA' to transform normal plant cells into a tumor.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. Which type of cell junction facilitates communication between cells by connecting the cytoplasm of adjoining cells for the rapid transfer of ions and small molecules?

(1) Tight junctions
(2) Adhering junctions
(3) Gap junctions
(4) Synaptic clefts

■■■

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